

ScreenshotChecker

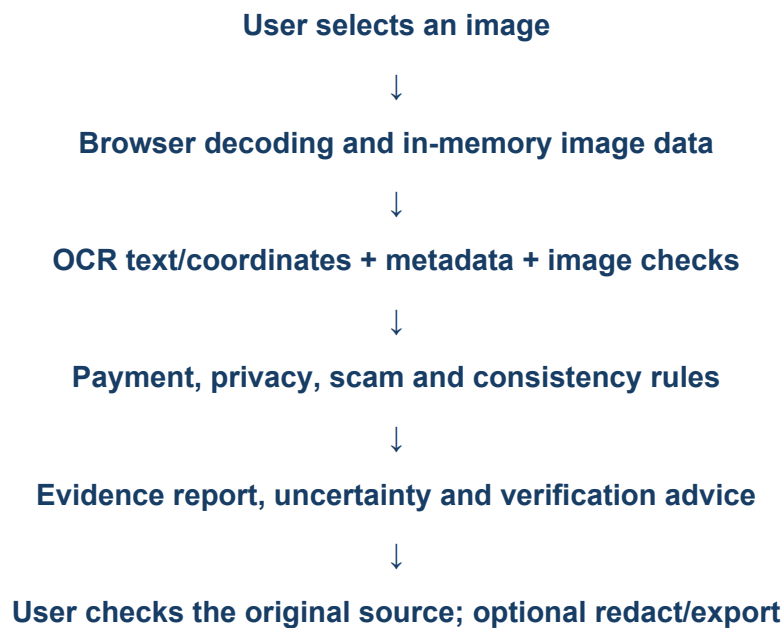
Technical Design & Evidence

Team CyberSuraksha | HSC|UP|00001 | 28 September 2026

1. Purpose and scope

A browser-based toolkit for reviewing UPI/payment screenshots and digital images, extracting text, inspecting metadata and privacy risks, and explaining when independent verification is needed. It supports Digital Safety & Cyber Fraud Awareness. It does not authenticate bank transactions.

2. Architecture and data flow



3. Implementation map

Interface: Astro, TypeScript, Tailwind CSS.

OCR: Tesseract.js and browser workers.

Metadata: ExifReader; C2PA library integration.

Image processing: browser Canvas.

Payment pipeline: `src/lib/paymentAnalyzer/paymentPipeline.ts`.

Shared modules: `src/lib/analyzer` and `src/lib/utlis`.

Setup, privacy and testing

4. Reproduction

Clone <https://github.com/viratchauhan/screenshotchecker>. Use the pinned Node version in .nvmrc (26.9.0) and the package manager listed in package.json (npm 11.19.1). Run npm ci, then npm run dev; open the local address printed by Astro. npm run build produces the static site; npm run preview serves the build.

5. Privacy boundary

Core OCR and image inspection run in the browser. OCR assets may require initial downloads. Optional external URL-reputation adapters and external reverse-image-search services must be treated separately from local image processing. The checked configuration is a static build; the presence of an adapter does not prove that its hosted endpoint is active.

Optional server adapters reference VT_API_KEY / VIRUSTOTAL_API_KEY, GOOGLE_WEB_RISK_API_KEY / GOOGLE_SAFE_BROWSING_API_KEY / GOOGLE_API_KEY, and URLHAUS_API_KEY / ABUSE_CH_API_KEY. URLHAUS_DISABLE=true disables that adapter. Keep credentials server-side; disclose any data sent to an enabled external service.

6. Samples and tests

Built-in sample generation: src/lib/samples/sampleData.ts. Reference patterns: src/data/fraud_sms_whatsapp_reference_dataset.txt. Tests exist for payments, OCR, metadata, redaction and report safety. Run npm test or npm run check; the latter runs tests and a build. The suite was inspected, not rerun during submission preparation.

7. Observed demonstration

On 28 September 2026 the built-in PhonePe example in the homepage integrated report returned INSUFFICIENT_EVIDENCE and extracted 284 characters. It recommended independent source verification. This single observed example is not an accuracy benchmark. The dedicated payment workflow is a separate module.

8. Risk controls and next validation

Recompression, cropping, low resolution, missing metadata and incomplete reference coverage can cause missed signals or false alarms. Preserve uncertainty; do not treat no-match as safe. Test labelled examples, benign edits and low-quality files. Measure false positives/negatives, task completion, device performance and users' understanding of source verification.

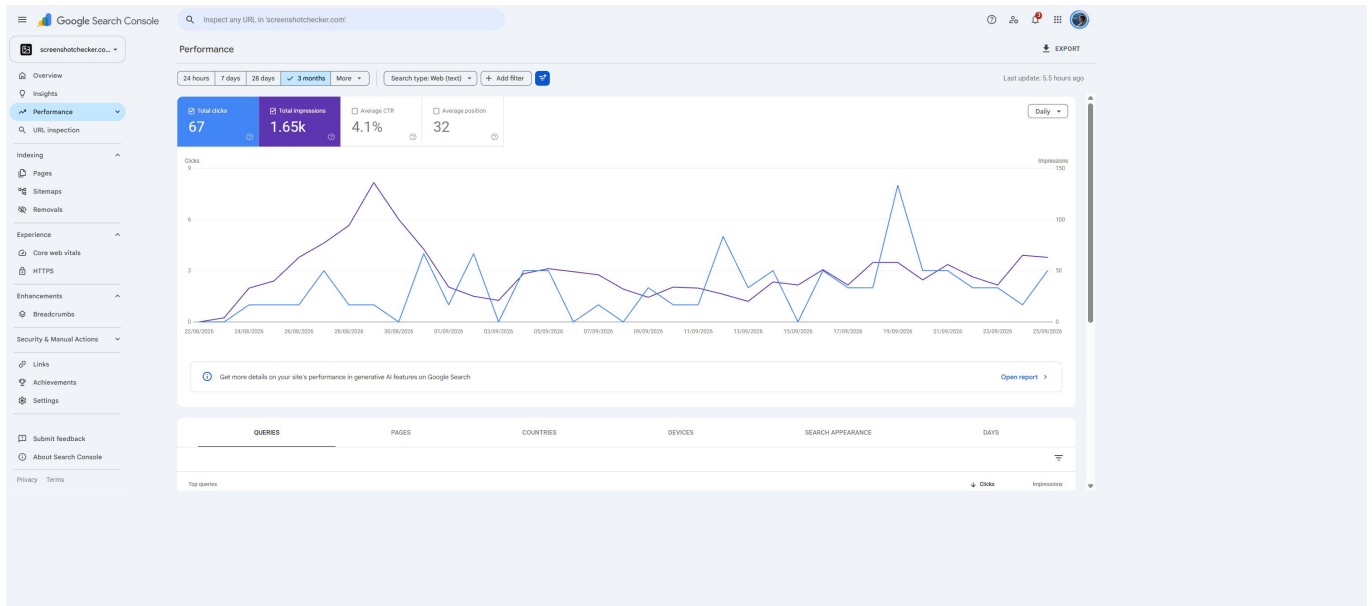
Early reach and intended impact

9. Search visibility evidence

67 clicks | 1.65k displayed impressions | 4.1% CTR | average position 32

Source: owner-authorized Google Search Console, screenshotchecker.com, read 28 September 2026.

Selected filter: three months, Web. Chart dates displayed: 22 August-25 September 2026. These metrics do not measure unique users, completed checks, accuracy or fraud prevented.



10. Social contribution and evaluation plan

Expected benefits include improved digital literacy, safer sharing and more careful payment decisions among students, small merchants and everyday users. Intended SDG alignment: SDG 4 (literacy), SDG 8 (safer commerce) and SDG 9 (accessible digital innovation). These are project-level intentions, not evidence of achieved SDG outcomes.

Proposed pilot: recruit consenting students and local merchants; use synthetic or consented samples; compare source-verification choices before and after a guided demonstration. Record task completion and misunderstandings, then improve the guidance. No partnerships or prevented-loss figures are claimed.

11. Sources and disclosure

<https://screenshotchecker.com>

<https://github.com/viratchauhan/screenshotchecker>

<https://sdgs.un.org/goals>

Official HSC presentation and declaration templates; existing team registration.

Developer disclosure: limited AI assistance, including ChatGPT for grammar correction. Submission text and video narration were prepared with AI assistance.